Outlook

[PLDI'26 SRC] Accepted submission #45 "K-Sentry: A Verified Order-Sensitive..."

From PLDI'26 SRC HotCRP <noreply-pldi26src@hotcrp.com>
Date Fri 4/24/2026 10:29 AM
To Ghimire, Sudrit <aui25@txstate.edu>

Dear author,

The PLDI 2026 Student Research Competition (PLDI'26 SRC) program committee is delighted to inform you that your submission #45 has been accepted to appear at the conference poster session. Congratulations!

* Title: K-Sentry: A Verified Order-Sensitive Telemetry Accumulator
* Author: Sudrit Ghimire (Texas State University)
* Site: https://nam04.safelinks.protection.outlook.com/?url=https%3A%2F%2Fpldi26src.hotcrp.com%2Fpaper%2F45%3F%3D&data=05%7C02%7Caui25%40txstate.edu%7Ccaaac70fd7d4484036e308dea2163c97%7Cb19c134a14c94d4caf65c420f94c8cbb%7C0%7C0%7C639126413578468904%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIwLjAuMDAwMCIsIlAiOiJXaW4zMiIsIkFOIjoiTWFpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=qaG6ljqNAr1e5HZY2frS1f3%2Bve%2FdQYepkO%2FDtYA1p3U%3D&reserved=0

Our expectation is that you update your extended abstract to incorporate feedback from the reviews you received. We will assign a member of the program committee (a "shepherd") who can help you with this process if you have any questions. The deadline to submit the updated camera-ready PDF is *May 15, 2026*.

We hope to see you at PLDI,
Gowtham & Alexa

Review #45A
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Overall merit
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3. Accept

Reviewer expertise
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2. Some familiarity

Technical Feedback

This paper describes a log accumulator that is sensitive to modifications in the log, ensuring that if a log is consistent with its checksum, then the log was unlikely to be tampered with. The paper is reasonably on topic for PLDI but might be more at home in a crypto or systems venue.

The origin of the accumulator is not explained until too late. Why is x_i q^i^2 mod p the right accumulator? Put another way, I can see that it works, but I wondered why i^2 is the right thing. I see this is explained later in L233, but I don't quite understand the tradeoffs quantitatively. Also, I'm not quite sure how resilient this mechanism is to various kinds of attacks: the paper discusses the probability of collisions at random (if I understood correctly), but I'm unclear how easy it would be for an attacker to modify the log in a way that still collides with the original.

The paper says that the code was verified with Verus, but lacking experience in Verus, I would have appreciated a hint as to what these theorems look like in Verus and what kind of proof was required. There's a summary around L200, but I'm curious what the actual code looks like.

The paper does not put the performance measurement in context. There are several other mechanisms summarized in Table 1; a performance comparison with those would be very helpful. For example, if this takes hardly any more computation than XOR, then it's clearly very cheap.

I wonder about the benefits of verification. Were any bugs found? Is there a reason to believe that the unverified algorithms are likely buggy? The fact that it was verified is viewed as an advantage (Table 1), so is there a reason we should use K-Sentry rather than verifying an existing algorithm (e.g. is verifying the other implementations infeasible)? Or is the main advantage of K-Sentry not that it is verified, but instead that it is order-sensitive and more compact (L224)?

Writing Feedback

I think the paper could do a better job of positioning. Why do we need a new log integrity mechanism with respect to practical needs? That is: the paper presents the technical shortcomings of prior approaches, but I'm unclear on whether these shortcomings present practical problems.

L173: I found the || operator unclear until I read the "why it matters" paragraph. Moving that paragraph earlier would help clarify, as would explaining the || operator explicitly. In general, I think putting the "why it matters" paragraph before the theorem statements might help.

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Review #45B
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Overall merit
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2. Unsure

Reviewer expertise
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2. Some familiarity

Technical Feedback

Summary
The paper proposes a novel telemetry algorithm over a finite field and describes a formally verified implementation of it.

Comments
At a high level I think this paper focuses too much on describing and justifying what seem to be fairly trivial theorems, and not enough time justifying the choice of hashing function. Does the quadratic weighting scheme it have particular properties that make it stronger than a traditional rolling hash? Is it somehow more amenable to verification? You briefly mention this in a single sentence in section 5, but I think these are more interesting questions worth spending time on; you definitely have the space for it.

You mention "verification overhead" as a reason your technique has lower throughput compared to standard rolling hashes, but surely the verification isn't happening at run-time, right?

Writing Feedback

Context
I think a formally verified implementation of something important like a hash function is reasonably well-motivated for a PL audience.

Technical presentation
As I've mentioned above, I think the paper spends too much time overexplaining the theorems while still missing important details. For example, what are the `nonzero_poly_has_few_roots` and `ramanujan_seq1/2/3` lemmas mentioned in section 3? These are named but as far as I can tell never defined.

I think the theorems presented in 3.1-3.4 are fairly obvious. Its definitely good to state them, but you don't need to spend so much time justifying why these are the theorems you chose to prove. In particular, I think the "_why it matters_" sections are unecessary and often kind of confusing; in particular, these read kind of LLM-generated to me.

Evaluation
Seeing throughput numbers is good, and I appreciate comparing to traditional rolling hash techniques (though again, I'm still curious why there should be any verification overhead). Along those lines, if a core contribution claim being made here is "this accumulator has better properties with respect to localizing errors", I'd expect to see some experiments showing its ability to detect multiple errors at once, etc.

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Review #45C
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Overall merit
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3. Accept

Reviewer expertise
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1. No familiarity

Technical Feedback

This paper presents K-Sentry which is a telemetry integrity mechanism. The goal is to have a mechanism that is streaming, order-sensitive and verified. The paper contrasts K-sentry with other telemetry integrity mechanisms. It explains the mathematical formula for K-sentry, and the key theorems involved in its verification. It concludes with an evaluation of the prototype.

Question: why does K-Sentry need to have a quadratic formula specifically vs a linear formula or a cubic formula? I assume there is a number theoretic reason, but it would be nice to know specifics. Also, though I am convinced the square gives you a good enough probability bound and other properties you need, would anything be improved with a higher power like a cube?

Writing Feedback

This paper would have benefitted from a longer introduction that explained the other telemetry approaches that K-sentry was being compared to more extensively (with citations where one could read more about other approaches would be helpful). For example, the table shows Merkle trees, but Merkle trees are not mentioned in the introduction. Also, when discussing the mathematical formula in the introduction, what are x, n, p and q? These variables are later defined in Section 2, but it is confusing in Section 1. The example in Section 2 was useful, as was the visualization and intuition behind the theorems. The evaluation was well explained.

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Review #45D
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Overall merit
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2. Unsure

Reviewer expertise
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3. Knowledgeable

Technical Feedback

The paper considers the problem of assigning a unique hash value to a sequence of values, for which it

defines a K-sentry accumulator which can computed in a streaming fashion and which also ensures that the hash value is sensitive to the order of values in the sequence. In addition, the implementation in Rust is also verified to satisfy basic functional properties.

While the paper seems like a good exercise in verification of a useful functionality which satisfies some nice properties and is also quite efficient, I am unsure whether there are any non-trivial research challenges involved in the process. The definition of the K-sentry accumulator is quite straightforward (basically a weighted average of values where the weights are derived by repeatedly squaring a base q), and it implementation as shown in Fig. 1 is very simple. The properties that are verified are also quite trivial, basically just stating that every step of the implementation is correct. As such, I do not see any specific verification challenges; this type of problem would be covered in a standard graduate course on program verification. The implementation can actually be improved a bit: in every iteration, it seems that the weight q^{i^2} is computed from scratch, but instead a running value can be maintained appropriately updated in each iteration to take advantage of the weights computed in the previous iterations.

It also seems to me this method of computing weights could quite quickly result in very large weight values for the later indices, which could result in very large hash values. Is there any specific reason why this particular method of computing weights was used?

Overall, I feel that while this is a good first step in verifying lightweight telemetry implementations, which hasn't been before, the paper is a bit lightweight in terms of contributions. There are other well-known telemetry implementations such as Rolling hash and Merkle tree which seem to satisfy the same properties as the proposed K-sentry. Maybe the authors can consider verification of these well-known techniques as well.

Writing Feedback

The writing is mostly OK. Some minorn comments:
- What is K in the word 'K-Sentry' refer to? I don't see any such notation being used anywhere in the paper.
- In the abstract and the introduction, symbols such as q and p are used without being defined, which comes later in Section 2. Please add the meanings of symbols before using them.

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